

Assignment Documentation

Factory Inventory & Production Management System (MVP)
Risk manager

This document contains the project risk management plan and a structured risk log derived from the implemented system, sprint records, deployment notes, and known final-stage issues.

1. Risk Management Plan

1.1 Risk Management Objective

The objective is to identify and control risks that could affect delivery, functional correctness, data integrity, deployment reliability, and academic submission quality of the project.

1.2 Risk Categories Used in This Project

- Technical risks: authentication, stock logic, FEFO allocation, report accuracy, and deployment behavior.
- Schedule risks: sprint slippage, unfinished UI coverage, and late rework caused by blocker discovery.
- Quality risks: documentation mismatch, broken commands, missing verification evidence, or inconsistent role records.
- Operational risks: incorrect environment configuration, missing user onboarding data, or confusion caused by non-project browser errors.

1.3 Assessment Model

- Probability is rated as Low, Medium, or High based on how likely the issue is to occur during development, demo, or deployment.
- Severity is rated as Low, Medium, or High based on impact to the working demo, data integrity, or submission quality.
- Each risk is assigned a mitigation strategy and an owner role responsible for monitoring or response.

1.4 Risk Response Strategy

- Avoid: reduce exposure through clearer scope boundaries and earlier verification.
- Mitigate: add tests, validation, CORS controls, route mapping, and documented operational steps.
- Share: spread monitoring responsibility across project, quality, scheduling, and startup roles where appropriate.
- Accept: record low-severity residual issues that are non-blocking but still relevant to future maintenance.

2. Risk Log

Risk ID	Risk Description	Probability	Severity	Mitigation / Response	Owner / Status
R-01	Firebase user exists but Mongo role record is missing, causing sign-in success but access failure.	Medium	High	Document onboarding steps; use Users page/admin seeding; validate OWNER_EMAIL and OWNER_FIREBASE_UID setup before demo.	Project manager / Monitored

Risk ID	Risk Description	Probability	Severity	Mitigation / Response	Owner / Status
R-02	Incorrect GRN, batch, or dispatch transaction handling could create stock mismatches.	Medium	High	Use validated routes, stock movement logging, and smoke-check full stock-in to dispatch flow after major changes.	Start up manager / Mitigated
R-03	FEFO allocation or batch traceability may be misunderstood or incorrectly demonstrated.	Low	High	Use dispatch page and reports traceability view in demo; keep delivery-note and stock-movement evidence available.	Risk manager / Monitored
R-04	Supplier, product, or user master data inconsistencies could block downstream workflows.	Medium	Medium	Use CRUD validation, seed data, and pre-demo checks for required master records.	Quality manager / Mitigated
R-05	Deployment fails due to missing env values, wrong API base URL, or incorrect Vercel routing.	High	High	Follow README and DEPLOYMENT guides; verify server and client vercel config and env names before release.	Start up manager / Mitigated
R-06	CORS preflight fails for the deployed client origin.	High	High	Use function-based allowlist, explicit OPTIONS handling, and exact CORS_ORIGINS values.	Start up manager / Closed
R-07	Nested serverless routes such as /api/orders/my return NOT_FOUND in production.	High	High	Keep explicit /api/(.*) mapping to api/[...path].ts in the server Vercel config.	Start up manager / Closed
R-08	Late-stage lint/test/documentation drift delays release.	Medium	Medium	Use build, test, and lint gates before sign-off; update docs immediately after deployment changes.	Quality manager / Closed
R-09	Browser extension JavaScript errors are confused with project bundle failures.	Medium	Low	Check filename/source in DevTools; document extension-script troubleshooting in deployment notes.	Risk manager / Accepted
R-10	Scope creep into non-MVP finance or analytics features reduces completion quality.	Medium	Medium	Hold scope to implemented modules and defer enhancements to future backlog.	Project manager / Mitigated

3. Monitoring and Escalation

- Review risk status at sprint checkpoints and before any release or submission milestone.
- Escalate High/High risks immediately when they block demo access, corrupt stock data, or invalidate documentation.
- Keep residual accepted risks visible in final documentation so reviewers understand what is and is not a project defect.

4. Conclusion

The risk-manager assignment deliverables are satisfied through a project-specific risk management plan and a maintained risk log that reflect the actual system and the real deployment issues encountered.